SOC-BASED LITHIUM-ION BATTERY CONTROL

Presented by Group 8

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INTRODUCTION

WHAT IS BATTERY STATE OF CHARGE?

Battery state of charge (SOC) is a normalized quantity between 0 and 1 that indicates the charge level in the battery at the moment. An SOC of 1 means the battery is fully charged, while an SOC of 0 means it is completely discharged.

SOC means State of Charge, which shows the remaining battery capacity compared to full capacity
It works like a fuel indicator for the battery

$$SOC(\%) = \frac{Q_{remaining}}{Q_{rated}} \times 100$$

Where:

- $Q_{remaining}$ = Remaining battery charge
- Q_{rated} = Total battery capacity

In today’s world, mobile phones have become an essential part of daily life, making battery performance and lifespan very important. Most smartphones use lithium-ion batteries, which are sensitive to charging habits. Frequent overcharging (charging up to 100%) and deep discharging (below 20%) can significantly reduce battery health and overall lifespan.

so to overcome this problems in the lithium ion battery which use in the mobile we took some parameters.

MOTIVATION

In modern electronic devices, especially smartphones, battery performance and lifespan are major concerns. Lithium-ion batteries tend to degrade faster when they are frequently overcharged to 100% or deeply discharged below 20%. However, most users are unaware of these effects and continue inefficient charging habits, which leads to reduced battery life and frequent replacements. This not only increases cost but also contributes to electronic waste. Therefore, there is a need for a smart charging system that can automatically maintain the battery within an optimal range of 20%–80% State of Charge (SOC). Such a system can improve battery health, enhance efficiency, and promote sustainable usage of electronic devices.

PROBLEM STATEMENT

This project designs a circuit and simple software logic to restrict mobile phone charging between 20% and 80% SOC. An alarm or indication system is developed to improve battery health and lifetime.

OBJECTIVE

- To design a circuit that controls battery charging limits
- To develop software logic using matlab simulink for maintaining SOC between 20%–80%
- To prevent overcharging and deep discharge of the battery
- To implement an indication/alarm system for SOC limits
- To improve battery life and efficiency
- To study state of charge (SOC) variation with respect to time

$$SOC(t) = SOC(0) + \frac{1}{C_{bat}} \int i(t)\, dt$$

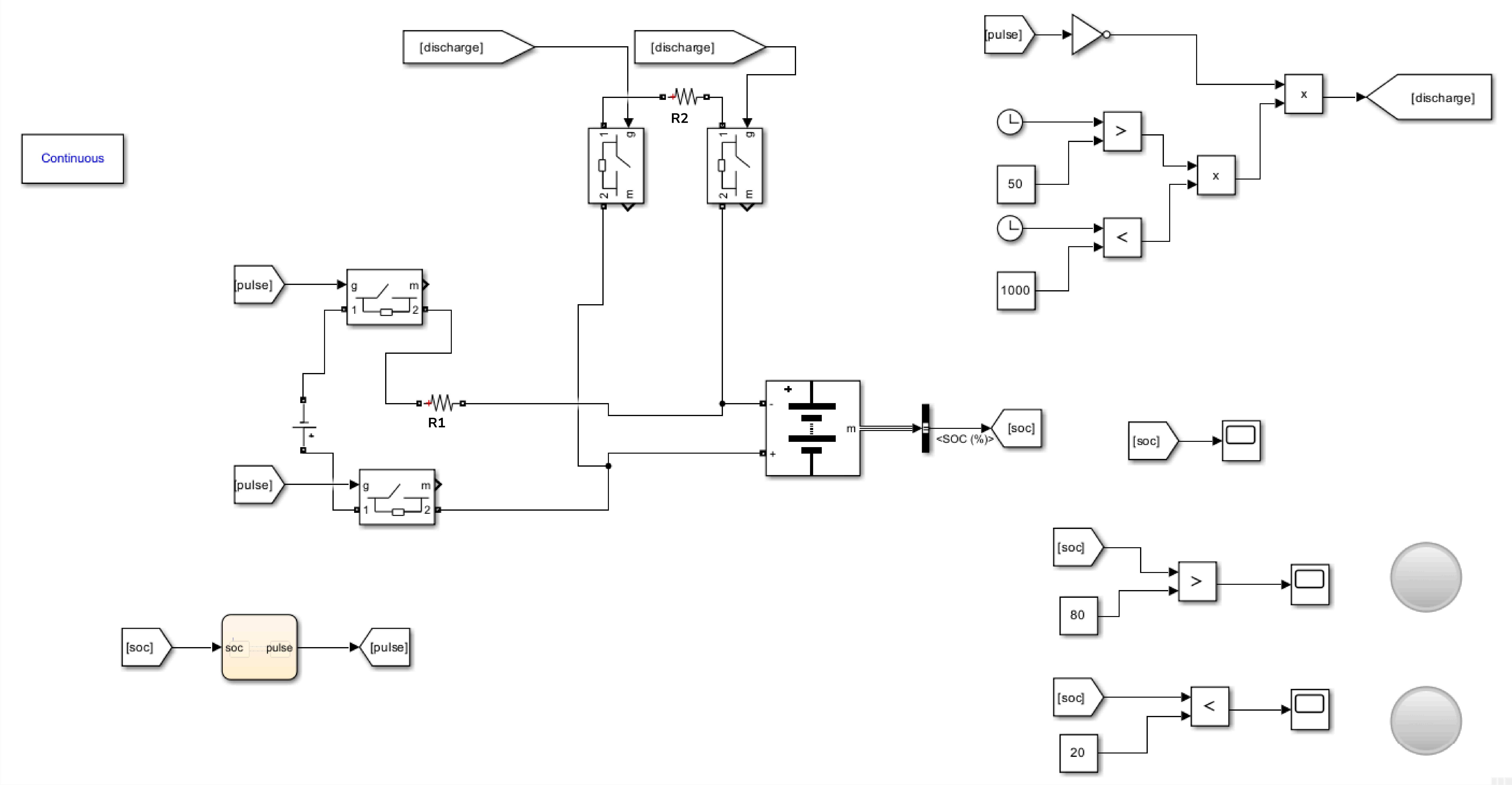
Where:

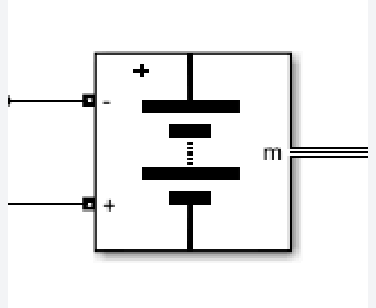
- $SOC(t)$ = state of charge at time t
- $SOC(0)$ = initial state of charge
- C_{bat} = battery capacity (Ah)
- $i(t)$ = battery current
- t = time

METHODOLOGY – SOFTWARE DESIGN

- SOC Calculation Algorithm
  Battery State of Charge calculated from Coulomb counting methods.
- Charging Control Logic
  System enables charging when SOC drops below 20% threshold. Disharging automatically enabled when SOC exceeds 80%.
- Alarm & Notification System
  lamp triggers at threshold boundaries. Visual LED indicators show current charging state.
- User Interface & Integration
  LCD display shows real-time SOC percentage. Status updates every 1s for responsive monitoring.

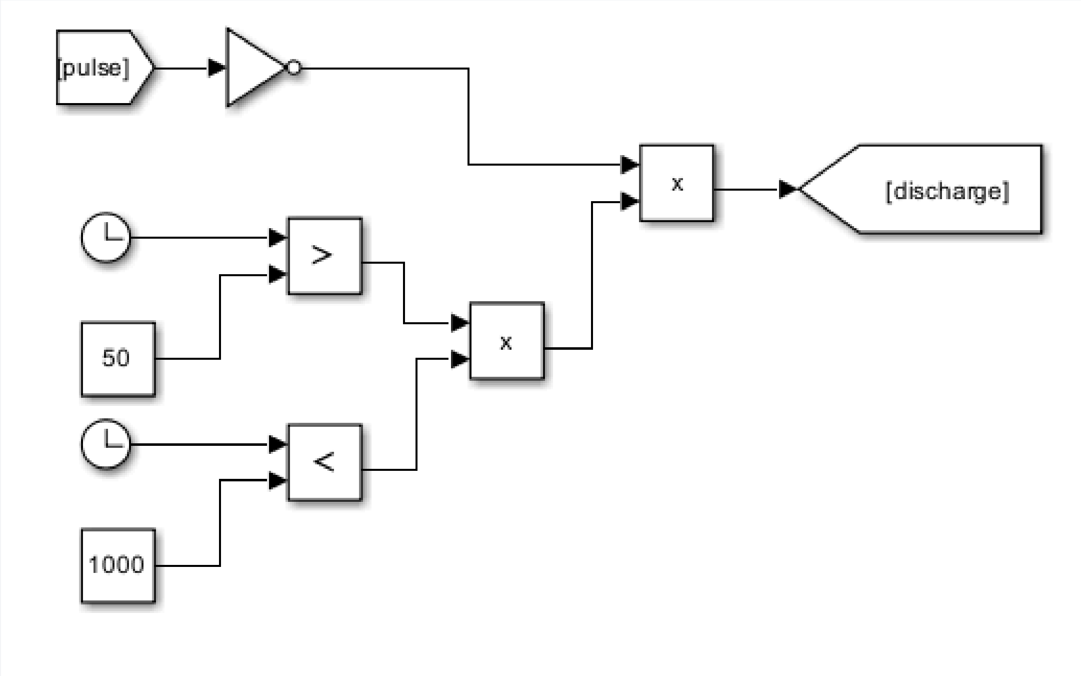
SIMULINK MODEL



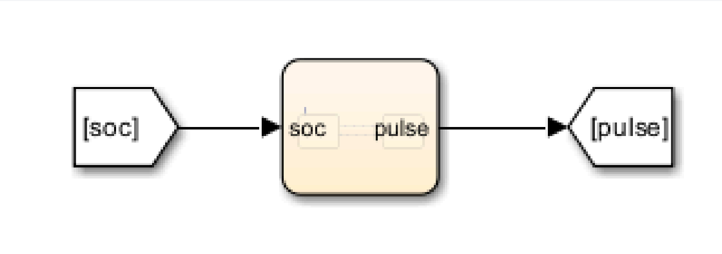


LITHIUM ION BATTERY

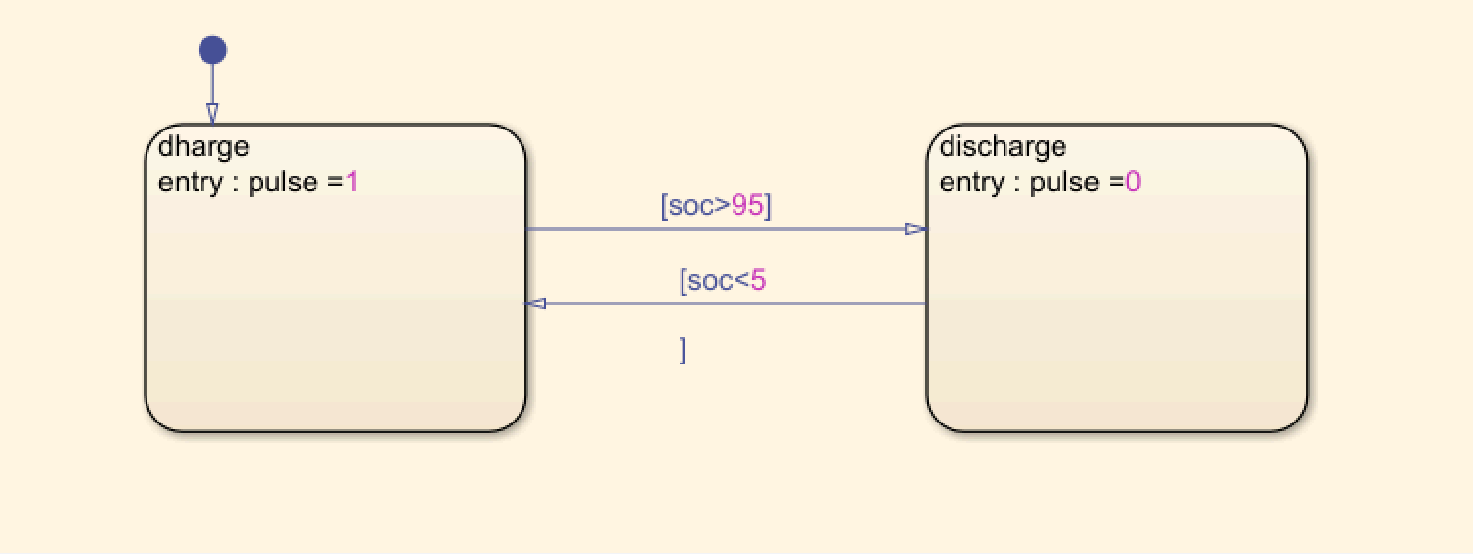
nominal voltage-7.2V
rated capacity-5.4ah



- this is the system makes the battery discharge. by the pulse from the pulse
- the battery will discharge with in the range mentioned with blocks
- if the battery is charged before the limit then it discharges in this interval



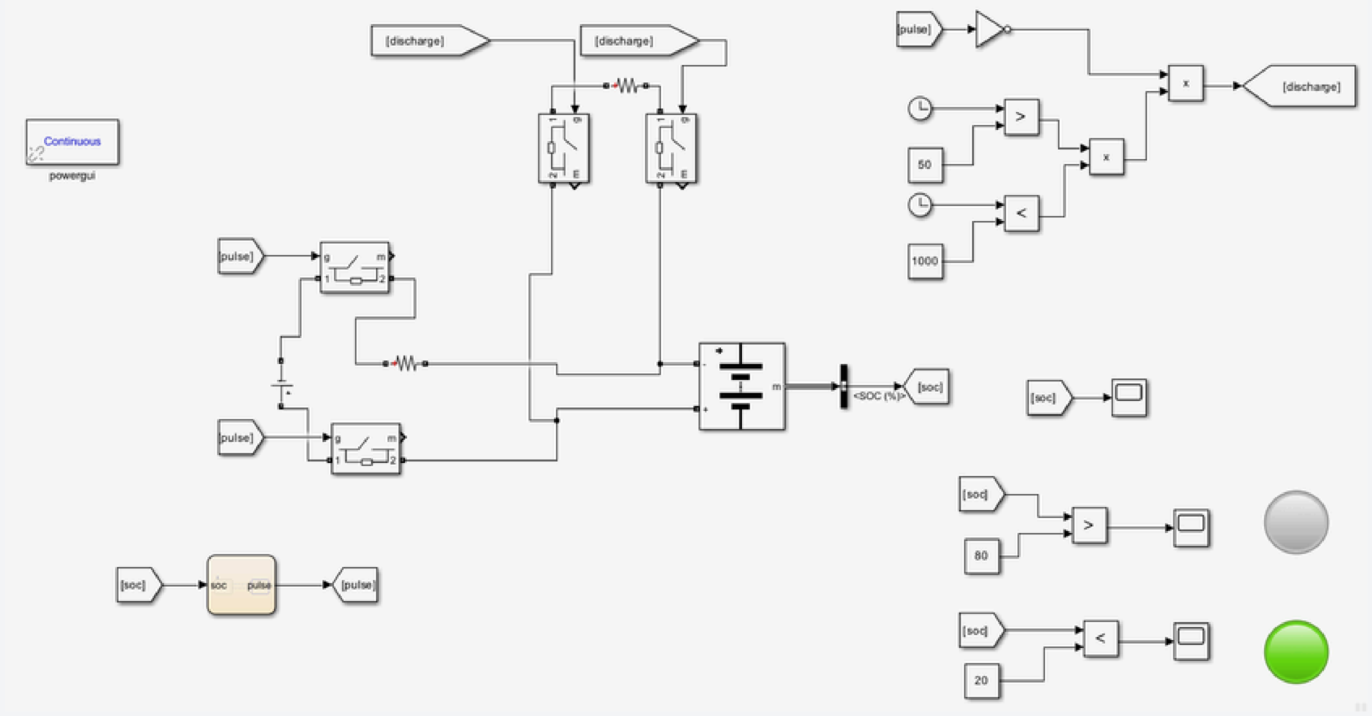
we used the chart block over the basic blocks like(switch,compare).



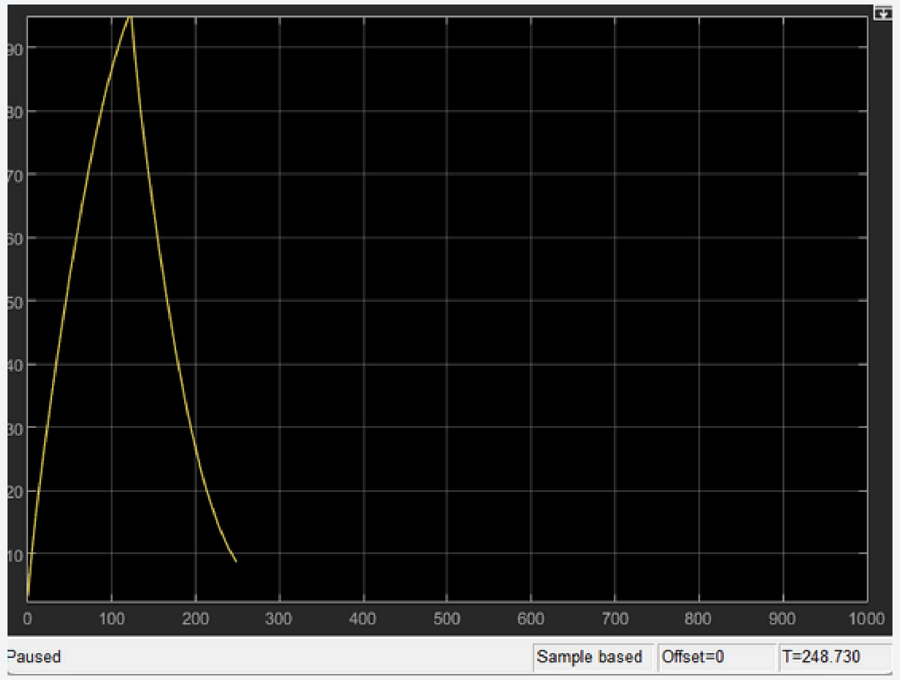
- decision-making logic (like a controller)
- state-based behavior (like ON/OFF, modes, conditions)

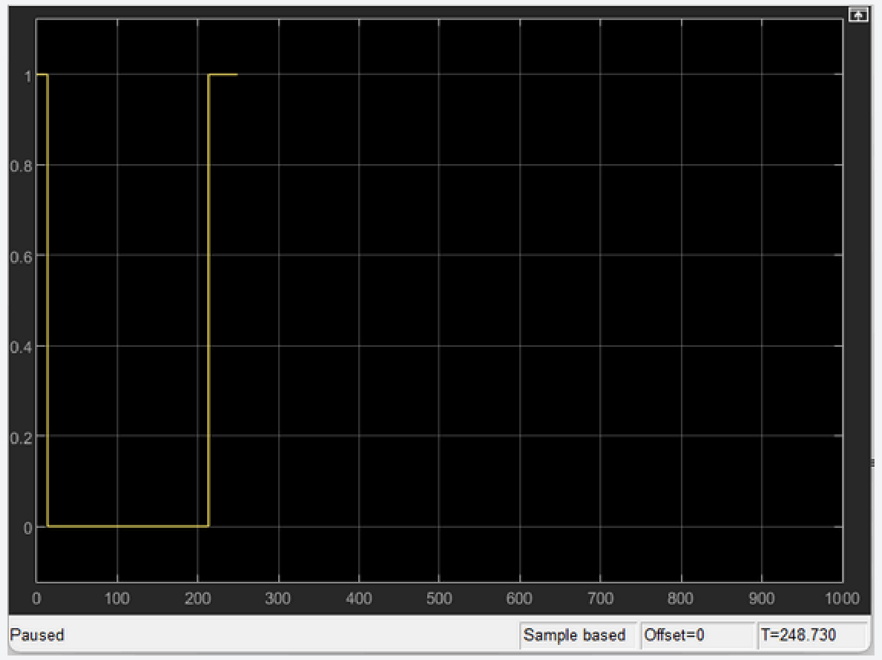
EXPERIMENTAL RESULTS

SOC<20

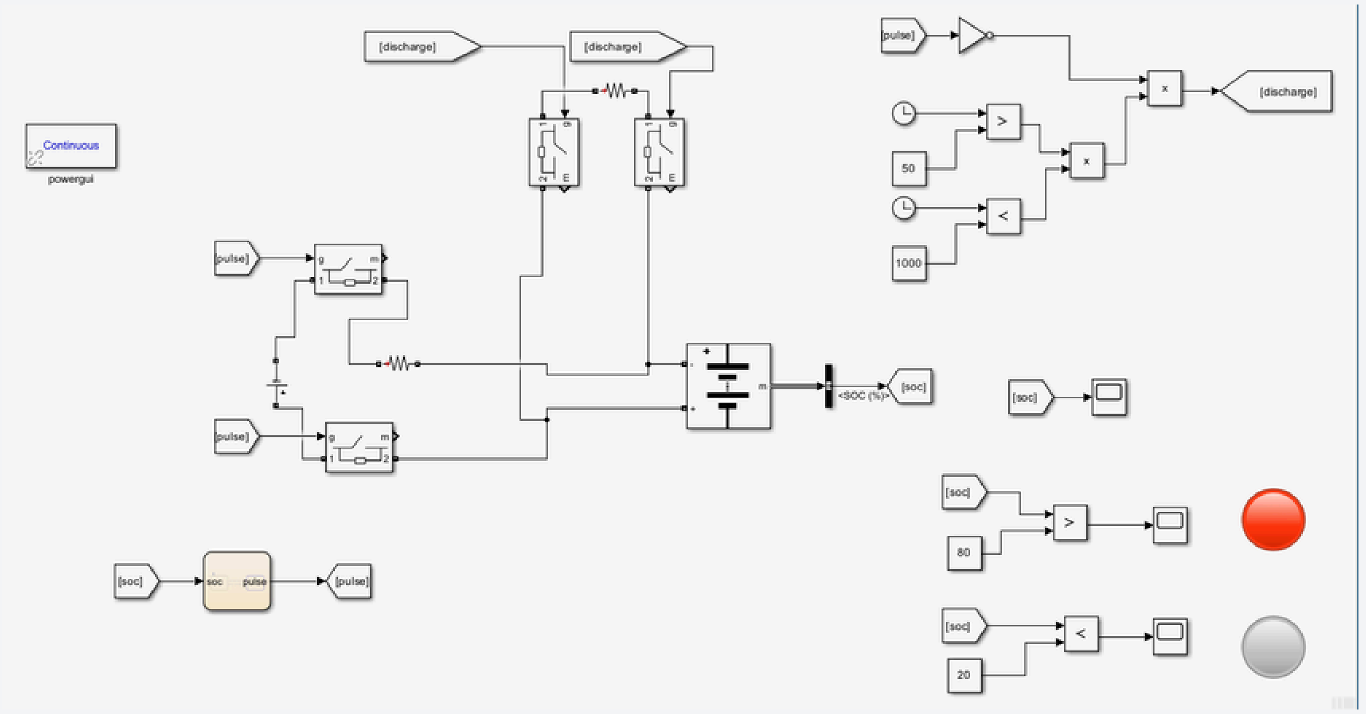


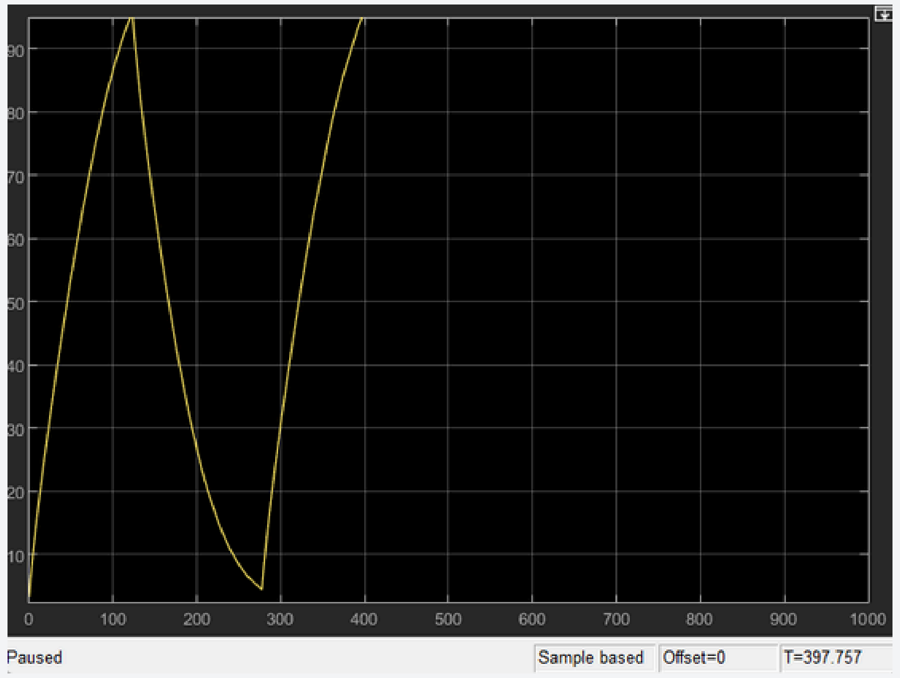
SOC (5%-95%)

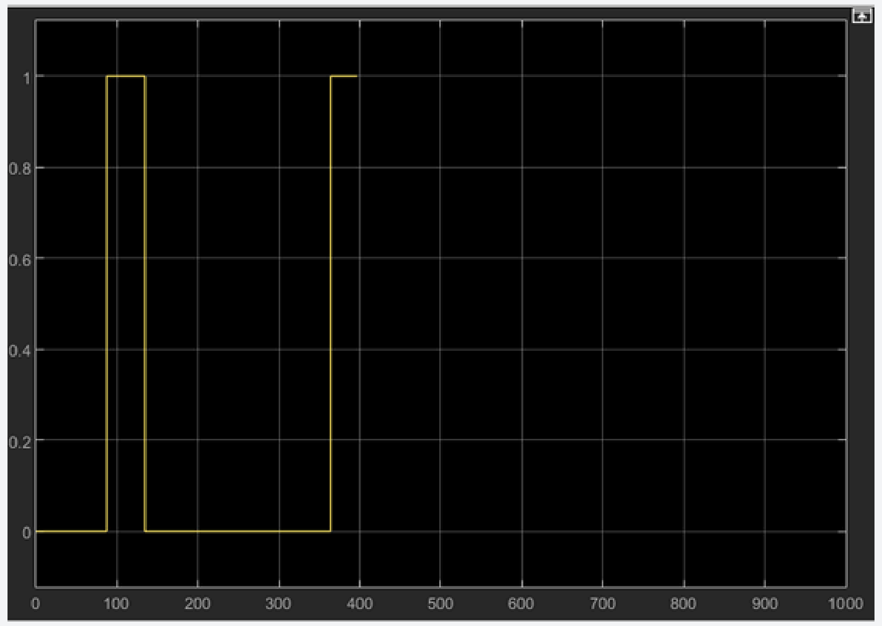




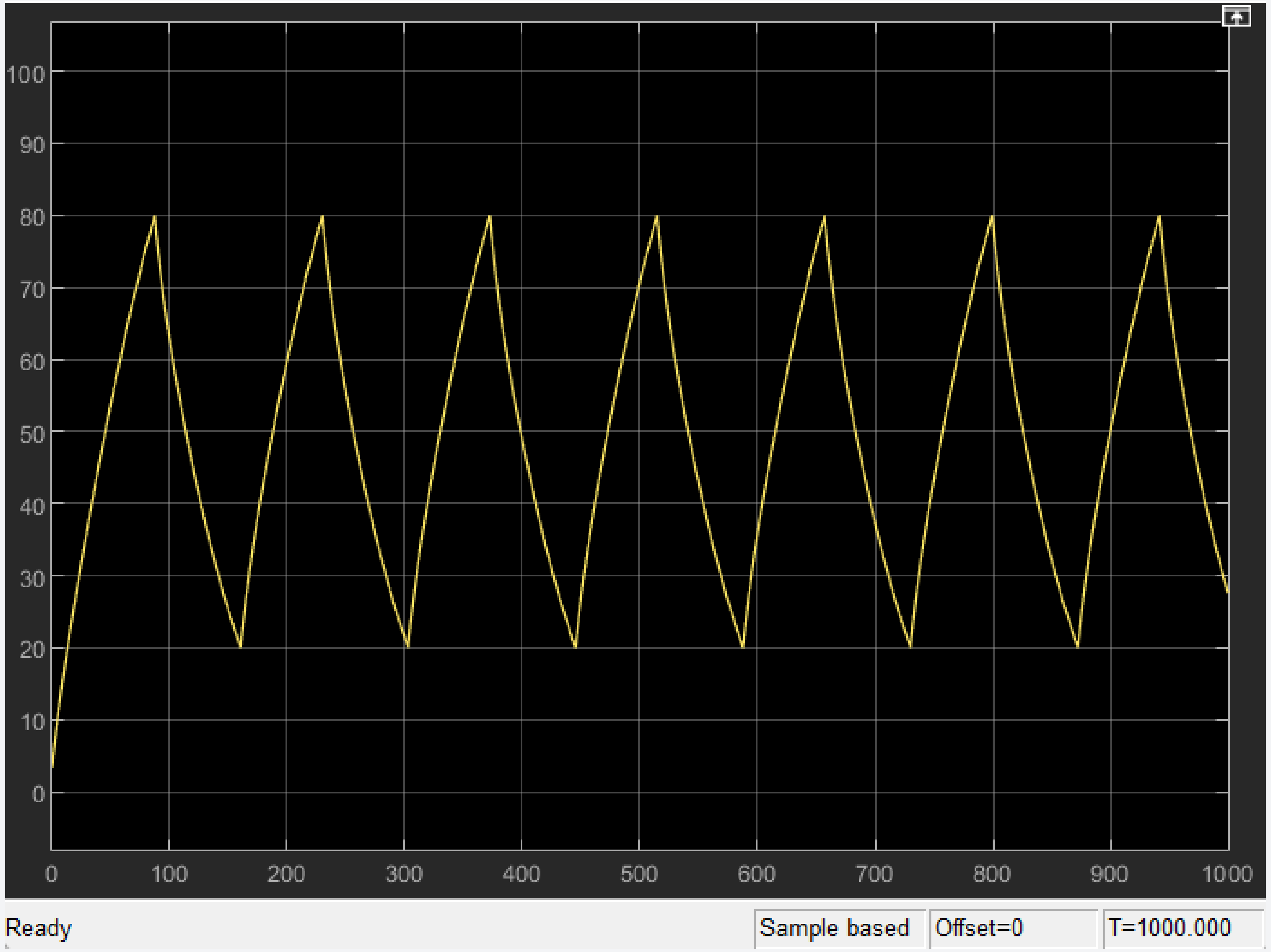
SOC>80





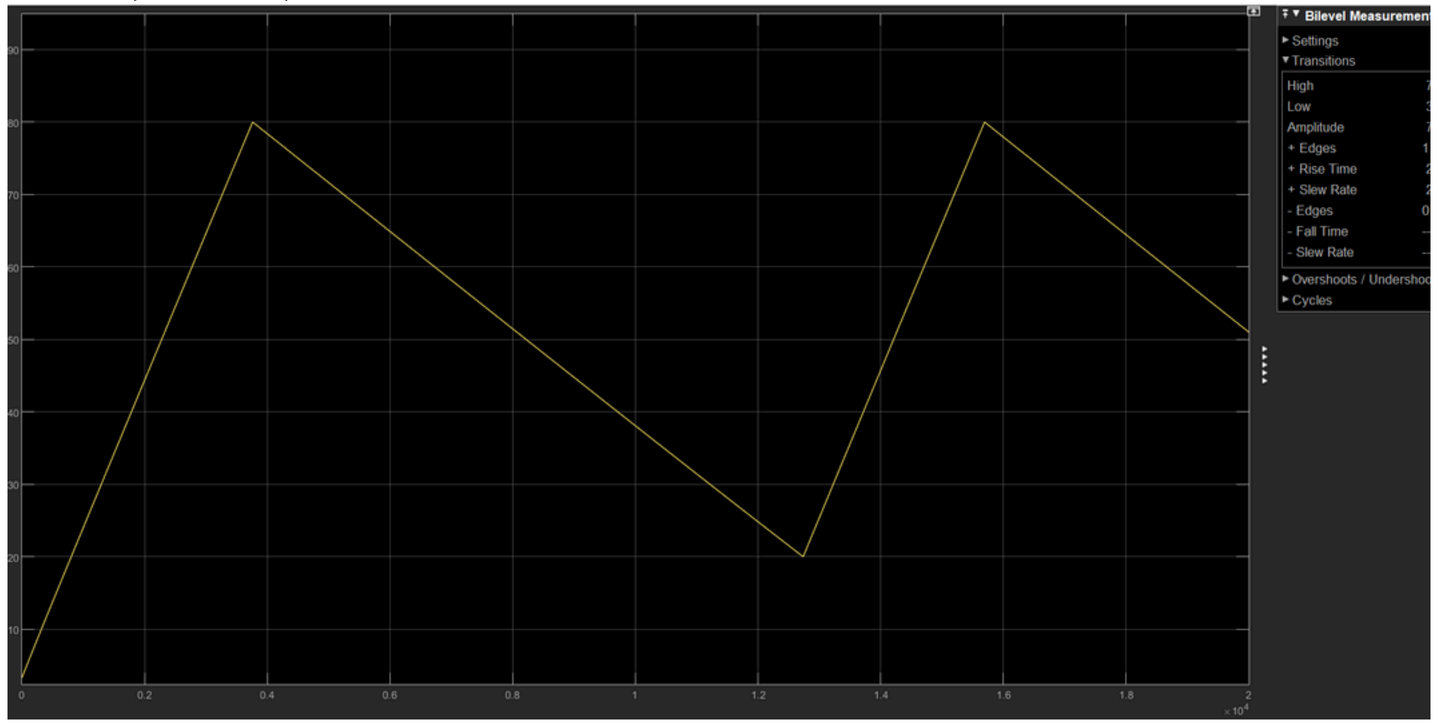


BATTERY SOC (20% TO 80%)

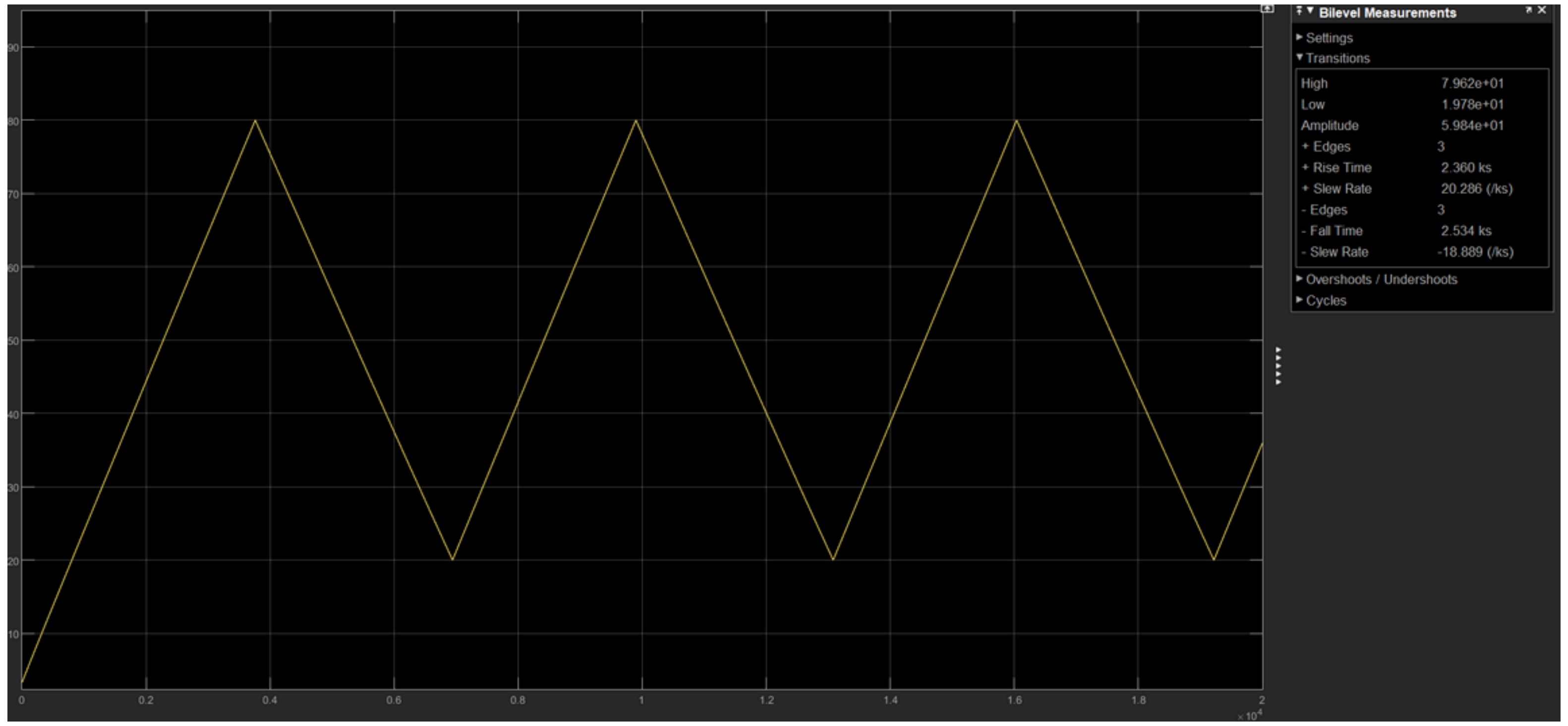


OBSERVATIONS

- P1(28W)
  P2(3.5W)



- P1(28W)
  P2(10W)



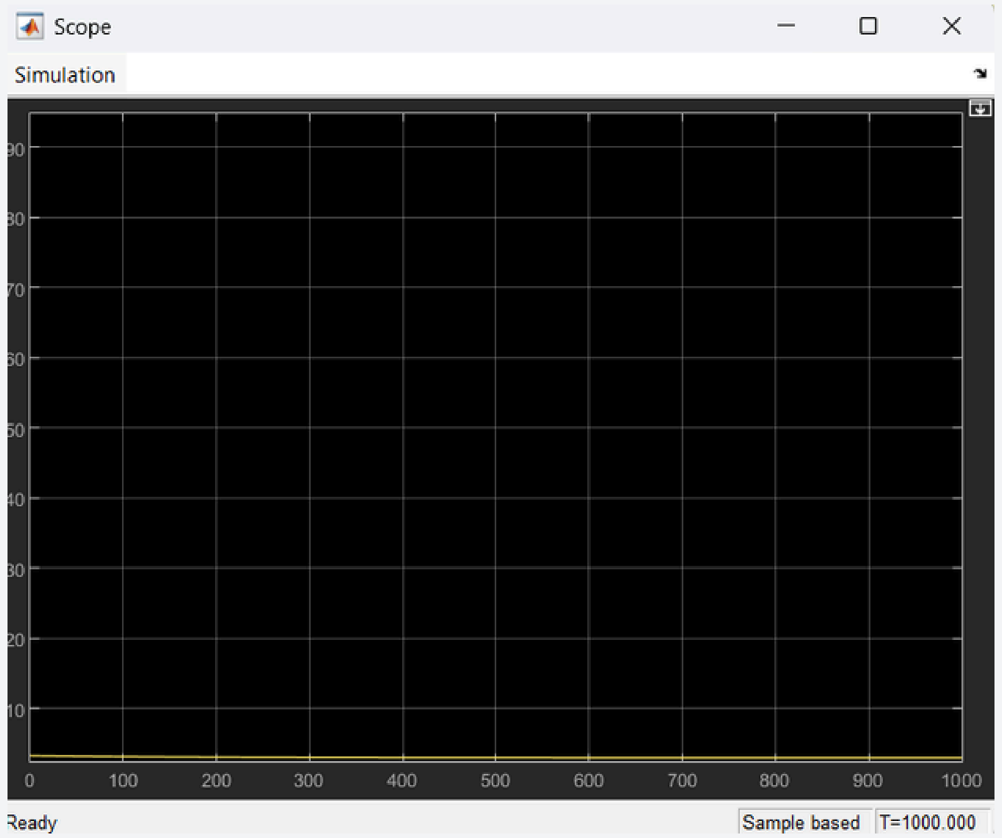
AT CONSTANT DC SOURCE

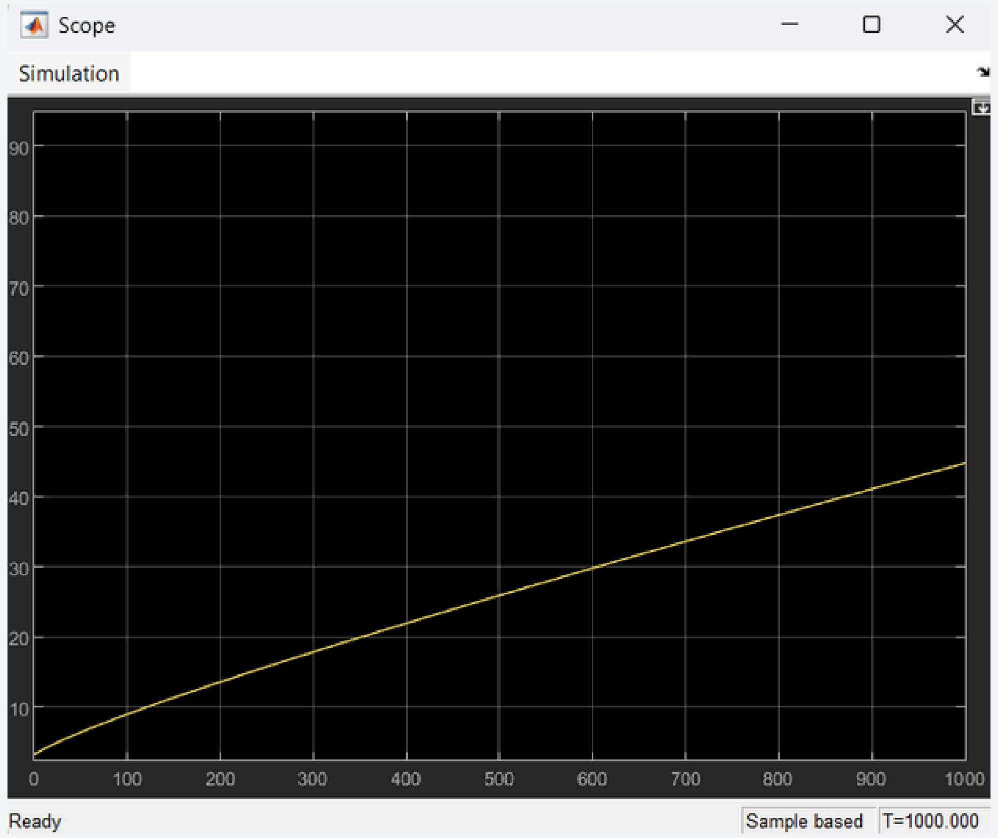
- High resistance → less current → less power
- Low resistance → more current → more power

- less current → less charge flow
- more current → more charge flow

Graph Interpretation

Battery nominal voltage: 3.85 V
Initial SOC = 0%
lower effective voltage(~3.2 V)
Source must overcome battery voltage to push current inside





Key Concept
A battery charges only when the external voltage exceeds its terminal voltage, enabling current flow into the battery.

Technical Relation
Higher voltage difference → Faster charging
Lower voltage difference → Slower charging

Important Note
Source voltage ≤ Battery voltage → No charging
This is why proper voltage control is critical in battery systems

Why Lithium-ion Battery

- High energy density
- Lightweight
- Fast charging
- No memory effect
- low self discharge
- Used in phones

Rated lithium ion battery for IPHONE

Parameter	Value (Approx)
Nominal Voltage	3.85 V
Max Voltage	4.4 V
Cut-off Voltage	3.0–3.2 V
Capacity	4400–4800 mAh
Energy	~17 Wh

FUTURE WORK

- Develop an advanced Battery Management System (BMS)
- Implement fast charging algorithms (CC–CV method)
- Add realistic battery modeling (internal resistance & nonlinear behavior)
- Include temperature and thermal effects
- Perform real-time hardware implementation (Arduino / embedded system)

CONCLUSION

- Successfully designed a system to control mobile charging between 20%–80% SOC
- Prevented overcharging and deep discharge of the battery
- Improved battery lifespan and efficiency
- Implemented an indication/alarm system for user awareness
- The project promotes better battery usage and sustainable practices

THANK YOU